

The OSI Model

The OSI model is a seven layer framework that describes how data moves across a network. The physical layer sends raw bits over a cable or radio link. The network layer chooses a path for packets between different networks. The transport layer provides end to end delivery between applications on two hosts. Encapsulation means each layer wraps the data from the layer above with its own header.

TCP and UDP

TCP is a connection oriented protocol that guarantees reliable, ordered delivery of data. Before sending data, TCP performs a three way handshake using SYN, SYN-ACK and ACK segments. UDP is a connectionless protocol that sends datagrams without guaranteeing delivery or order. Video calls and online games often use UDP because low latency matters more than retransmitting lost packets.

IP Addressing and Subnets

An IP address is a numeric label that identifies a device on a network. IPv4 addresses are 32 bits long, while IPv6 addresses are 128 bits long. A subnet mask separates the network portion of an address from the host portion. NAT lets many devices on a private network share a single public IP address.

DNS Resolution

DNS is a distributed system that translates domain names into IP addresses. A recursive resolver queries root servers, then top level domain servers, then the authoritative name server for the domain. Resolvers cache answers for the time set by the TTL value, which reduces lookup latency and load on name servers.

Routing

A router is a device that forwards packets between networks using a routing table. Distance vector protocols share routing tables with neighbours, while link state protocols flood information about links so every router can compute shortest paths. Dijkstra's algorithm computes the shortest path from one router to every other router in a link state network.